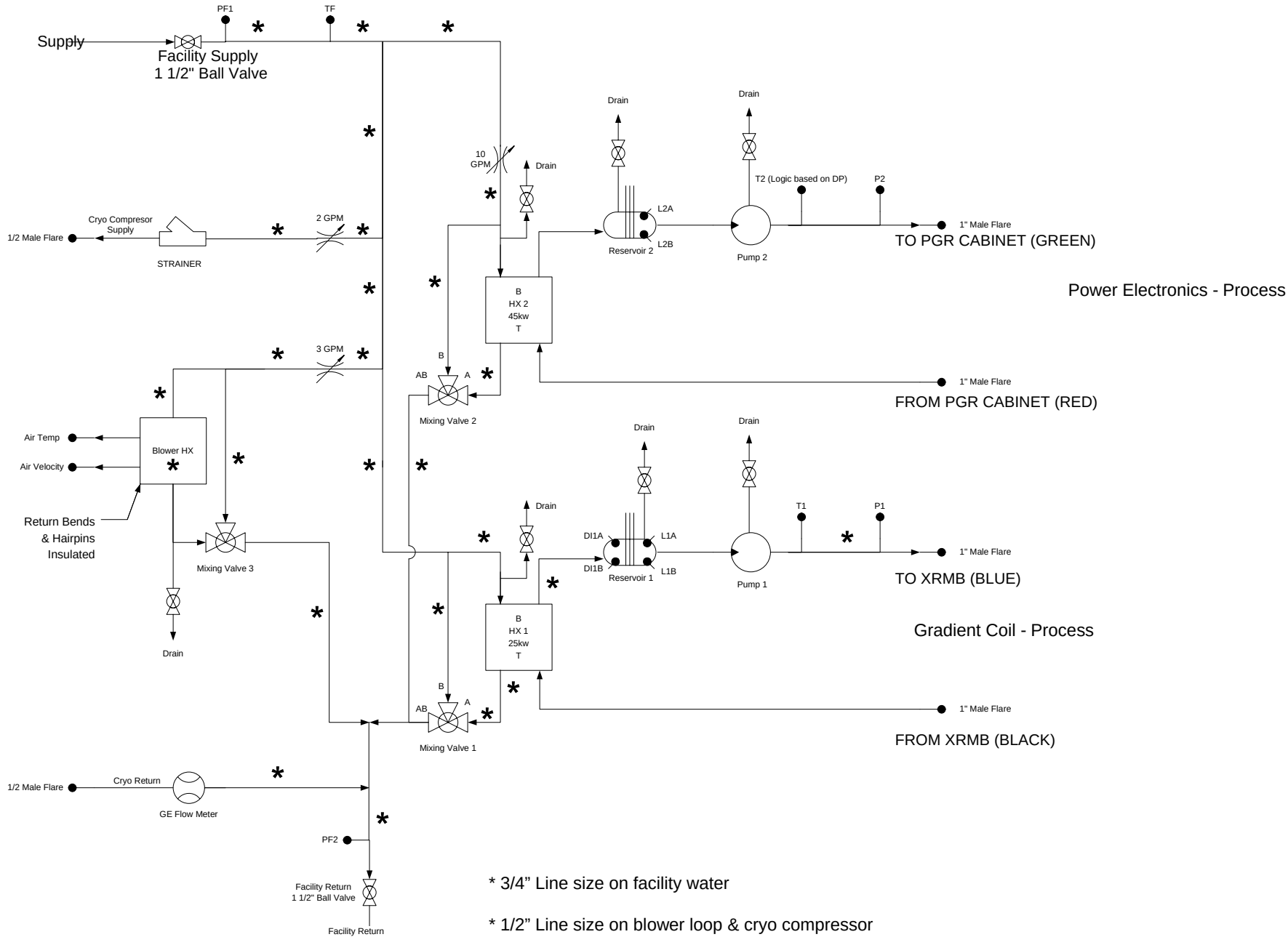


COOLING

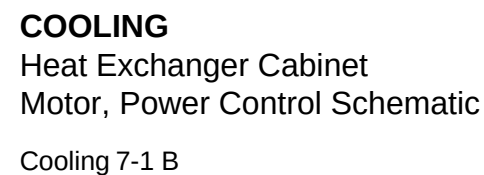
TABLE OF CONTENTS

<u>SECTION</u>	<u>PAGE</u>
7 – COOLING	
Heat Exchanger Cabinet.....	7-1
PLC Signal Boc.....	7-2
Air Flow and Cooling.....	7-3



- | | | |
|------|---|------------------------------------|
| * | = | Insulated Lines |
| PF1 | = | Facility Pressure Sensor Supply |
| TF | = | Facility Temperature Sensor Supply |
| PF2 | = | Facility Pressure Sensor Return |
| T2 | = | PE Temperature Sensor Supply |
| P2 | = | PE Pressure Sensor Supply |
| L1A | = | GC Level Sensor 1 |
| L1B | = | GC Level Sensor 2 |
| L2A | = | PE Level Sensor 1 |
| L2B | = | PE Level Sensor 2 |
| DI1A | = | GC Conductivity Sensor 1 |
| DI1B | = | GC Conductivity Sensor 2 |
| T1 | = | GC Temperature Sensor Supply |
| P1 | = | GC Pressure Sensor Supply |

- * 3/4" Line size on facility water
- * 1/2" Line size on blower loop & cryo compressor
- * Process drain to be through pumps & tanks
- * Facility to drain through HE's

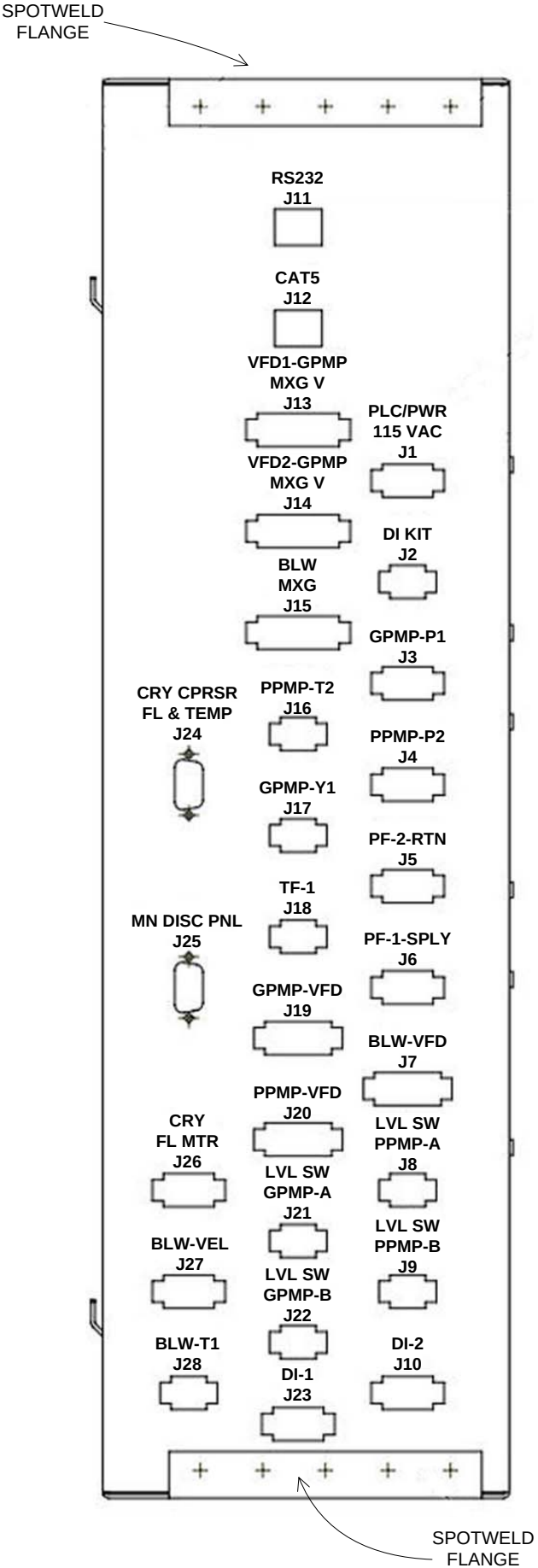




Heat Exchanger Cabinet Signal Box Diagram

Cooling 7-1 C

P# = Pressure Sensor
T# = Temperature Sensor



COOLING
PLC Signal Box

Bulkhead Label		Details
J-number	Description	
J1	PLC/PWR 115 VAC	Incoming signal box power from power box
J2	DI KIT	Provides 24V signal to relay control for 120VAC power to DI service outlet at top of power box
J3	GPMP-P1	Coolant pressure sensor on the gradient coil supply line
J4	PPMP-P2	Coolant pressure sensor on the power electronics supply line
J5	PF-2-RTN	Facility coolant return pressure
J6	PF-1-SPLY	Facility coolant supply pressure
J7	BLW-VFD	Blower variable frequency drive control/communication
J8	LVL SW PPMP-A	Power electronics reservoir liquid-level warning switch; upper location on reservoir, L2A
J9	LVL SW PPMP-B	Power electronics reservoir liquid-level alarm switch; lower location on reservoir, L2B
J10	DI-2	Gradient coil coolant resistivity sensor 2; sensor located in reservoir
J11	RS232	Serial communication to PLC from PGR J3
J12	CAT5	Ethernet communication to PLC from PGR J6
J13	VFD1-GPMP MXG V	Gradient coil plumbing loop mixing valve control
J14	VFD2-PPMP MXG V	Power electronics plumbing loop mixing valve control
J15	BLW MXG V	Blower plumbing loop mixing valve control
J16	PPMP-T2	Power electronics coolant temperature sensor in supply line
J17	GPMP-T1	Gradient coil coolant temperature sensor in supply line
J18	TF-1	Facility coolant supply temperature
J19	GPMP-VFD	Gradient coil variable frequency drive control/communication
J20	PPMP-VFD	Power electronics variable frequency drive control/communication
J21	LVL SW GPMP-A	Gradient coil reservoir liquid-level warning switch; upper location on reservoir, L1A
J22	LVL SW GPMP-B	Gradient coil reservoir liquid-level alarm switch; lower location on reservoir, L1B
J23	DI-1	Gradient coil coolant resistivity sensor 1; sensor located in reservoir
J24	CRY CPRSR FL & TEMP	Cryogen compressor flow rate and temperature output to Magnet Monitor; temperature is facility coolant supply temperature
J25	MN DISC PNL	Output signal (currently not used) capable of shutting down the system at the main disconnect panel in the event of a major loss of coolant detected.
J26	CRY FL MTR	Cryogen compressor coolant flow rate meter
J27	BLW-VEL	Air blower velocity input
J28	BLW-T1	Air blower temperature sensor

